

AMERICAN ELECTRIC POWER DATA GOVERNANCE PRINCIPLE



DATA GOVERNANCE GUIDING PRINCIPLES (DGGP)

EXECUTIVE SUMMARY

American Electric Power operates 40,000 miles of high-voltage transmission across 11 states, serving 5.6 million customers through six independently managed operating companies. This geographic and organizational scale has historically produced inconsistent data definitions, duplicate records, and manual reconciliation workflows that undermine enterprise-wide decision-making and exposing AEP to regulatory risk under NERC CIP and applicable state utility commission standards.

These ten guiding principles constitute AEP's enterprise data constitution: the permanent framework within which all data-related decisions, systems, and projects must operate, regardless of which business segment or operating company is involved. They are not project-specific guidelines. They are organizational commitments that govern how AEP collects, classifies, accesses, protects, and uses data across Transmission, Distribution, Generation, and all Vertically Integrated Utility operations.

The framework adopts a federated governance model: an enterprise layer sets standards and accountability structures, while each operating company retains stewardship of its regional data assets within those standards. This structure balances the consistency required for enterprise-wide analytics with the operational flexibility demanded by 11 different state regulatory environments. Together, the ten principles form a coherent system where data quality, security, accessibility, and education reinforce each other, creating the foundation AEP needs for reliable, data-driven operations through its \$54 billion capital investment horizon

1. Data Governance Strategic Approach & Organization

AEP has six different operating companies that work in 11 states across the eastern and central United States. Each company has historically managed its data. This fragmentation creates significant operational challenges across all business segments, including Transmission, Distribution, Generation, and Vertically Integrated Utilities (VIU). Analysts had to go through many systems, fix inconsistent data labels, and manually settle conflicting definitions before they could obtain a unified customer or operational view. Without a structured governance framework, data inconsistencies compound over time, undermining the reliability of enterprise-wide decision-making across AEP's entire organization.

To address this, AEP will adopt a federated governance model consisting of two coordinated layers: an enterprise governance layer and an operating company layer. At the enterprise level, a Chief Data Officer (CDO) leads a central governance body responsible for establishing company data standards, policies, and accountability frameworks that apply uniformly across all operating companies and business segments. At the operating company level, dedicated governance teams execute within those enterprise standards while maintaining ownership of their regional data assets. This dual-layer structure ensures both consistency and operational flexibility across AEP's geographically distributed organization.

Four roles anchor this governance structure at both levels. Data Stewards are responsible for defining data policies and providing implementation guidance to ensure those policies are applied correctly across all business units. Data Owners define data requirements for their respective domains and are accountable for ensuring data compliance with regulatory and organizational standards. Data Architects design and maintain the technical infrastructure that supports data storage, integration, and security across the entire enterprise. Data Users are business and operational staff who use data daily and are responsible for ensuring they adhere to the established governance policies in their work. By embedding all four roles at both levels, AEP ensures that governance is not a centralized bottleneck but a shared responsibility that scales consistently across the company.

2. Process Management Approach

AEP operates in a continuously evolving energy environment, where a \$54 billion capital investment plan is introducing new grid technologies, renewable energy assets, and digital systems at a rapid pace across all business segments. Each new system brings new data sources in potentially different formats, and without a structured process to absorb and standardize these changes, AEP risks returning to the same fragmented, inconsistent data that governance is designed to eliminate. Failure to keep pace with change does not simply slow progress; it exposes AEP to regulatory fines, unreliable operational data, and erosion of the data trust that enterprise-wide decision -making depends on.

To prevent this, AEP's enterprise governance team will conduct monthly process reviews to assess whether existing governance policies, data standards, and procedures remain effective and current across all parts of the organization. These reviews will incorporate developments in regulatory requirements, particularly NERC CIP standards, as well as emerging industry best practices in data management. When gaps or outdated practices are identified, the governance team is responsible for updating policies and communicating changes to Data Stewards and Data Owners across all operating companies and business segments in a timely manner. This continuous improvement cycle ensures that governance evolves alongside the organization rather than becoming a static framework that loses relevance over time.

Technology-driven risk is a central concern in AEP's process management approach. Its operations across generation, transmission, distribution, and customer services depend on the integrity of data flowing from SCADA systems, Advanced Metering Infrastructure (AMI) networks, and financial systems. If governance processes break down and data quality deteriorate, the consequences extend beyond reporting errors. Delayed or missed detection of grid failures can result in widespread power loss, safety incidents, and significant penalties under federal reliability standards. AEP will therefore integrate data governance risk assessments into its enterprise risk management framework, with automated monitoring tools providing real-time alerts when data pipelines or quality metrics fall outside acceptable thresholds, enabling the governance team to intervene before operational impact occurs across any part of the business.

KPI

MEASURE	TARGET	RATIONALE
Monthly policy reviews	12 completed per year	Ensures governance evolves at the same pace as AEP's capital investment programme

Policy change communication	5-day SLA to all operating companies	Prevents operating companies from unknowingly operating under outdated standards
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3. Data Accessibility

AEP employs a diverse workforce spanning field technicians, data analysts, customer service representatives, financial professionals, and executive leadership across all 11 states and business segments. Each employee group requires access to different data to perform their responsibilities effectively. A field technician needs real-time device health metrics and sensor readings relevant to their service territory; they have no operational need for corporate financial records. Executives, by contrast, require visibility across all data domains to make informed strategic decisions for the entire enterprise. Granting blanket access to all employees creates unnecessary security exposure, while overly restricting access slows operations and frustrates users across the organization. AEP's accessibility principle is based on a clear need-to-know standard: every employee can only see the data that is necessary for their job, no matter which business segment or operating company they work for.

Access decisions at AEP will be made collaboratively by technical leads, Data Owners, and the governance team, with legal and compliance input where confidentiality obligations apply. This multi-stakeholder approval process ensures that access grants are both operationally appropriate and regulatory compliant across all of AEP's business units. Access permissions will be formally reviewed whenever an employee changes roles and immediately revoked upon departure from the organization. To ensure no employee can claim they were unaware of their access boundaries, mandatory data access training will be delivered during onboarding and refreshed on a regular basis, establishing clear and documented awareness of what each employee is and is not permitted to access.

All data access events across AEP's systems will be recorded in a centralized audit log, giving the organization the visibility needed to detect unauthorized access after the fact. When a potential violation is identified, a dedicated security investigation team will conduct a confidential review, notifying the employee's direct supervisor while ensuring the matter is handled discreetly to protect the individual's dignity. Outcomes will be proportionate to the severity of the violation, with minor breaches resulting in mandatory retraining and serious or intentional violations escalating to termination. This balanced approach ensures accountability without creating a culture of fear around data access across any part of the company's operations.

KPI

MEASURE	TARGET	RATIONALE
Access revocation on departure	100% revoked within 24 hours	A former employee retaining access to customer PII or grid data for even 48 hours creates a measurable and auditable risk window
Role-change access reviews	100% reviewed within 5 business days	Prevents privilege accumulation as employees move between roles across the six operating companies

4. Data Security

AEP is a key provider of national infrastructure that serves 5.6 million customers in 11 states. This makes its data assets a high-value target for cybersecurity threats in all business areas. External actors may attempt to infiltrate grid control systems, customer databases, generation management platforms, or financial records through sophisticated cyberattacks targeting any part of the enterprise. However, the most significant vulnerability in any organization is human: employees across all levels and locations remain susceptible to social engineering, phishing, and inadvertent data exposure through careless handling of sensitive information. AEP's data security principle therefore addresses both the technical and human dimensions of data protection, recognizing that no technical control is sufficient without a workforce that understands and actively supports security practices throughout the entire organization.

AEP will implement a data classification system categorizing all data into three tiers: public, private, and confidential. Public data includes general corporate information suitable for external disclosure. Private data includes internal operational and employee records accessible only to authorized staff. Confidential data includes SCADA control system data, customer personally identifiable information (PII), generation asset configurations, and NERC CIP-regulated assets, all of which will be subject to the most stringent controls, including end-to-end encryption, multi-factor authentication, and strict network segmentation. Regular vulnerability assessments and penetration testing will be conducted across all systems to identify and remediate weaknesses before they can be exploited, and all security patches will be applied within defined remediation timeframes.

In the event of a data breach, AEP will follow a documented incident response procedure with NERC CIP compliance requirements embedded directly into each stage. Upon discovery, the incident will be immediately contained and escalated to the dedicated security response team, with regulatory notifications issued within the timeframes mandated by NERC CIP and applicable state regulations. AEP is committed to transparent reporting: while regulatory fines may result, proactive disclosure is far preferable to concealment, which carries significantly greater legal and reputational consequences for the entire enterprise. Following containment, a thorough root cause analysis will be conducted and all identified vulnerabilities patched before affected systems are restored to operation, ensuring the same breach vector cannot be exploited again anywhere across AEP's infrastructure.

Four-tier data classification

Every data asset at AEP is assigned to one of four classification tiers. The tier determines who can access it, what controls protect it, and what approval is required to gain access. Controls increase with the consequence of unauthorized exposure, not with operational convenience.

Tier 1: Public

Information suitable for external disclosure without restriction. No harm to AEP or its stakeholders if released publicly.

Examples: annual reports, press releases, general service territory maps, published rate schedules

Access: all employees and public. No controls required.

Tier 2: Internal

Operational and business information for internal use only. Unauthorized disclosure could cause operational disruption or competitive disadvantage.

Examples: internal process documentation, employee directories, maintenance schedules, non-public financial projections

Access: authenticated employees only. RBAC enforced. Head of team or department approves. Every access logged with timestamp and user ID.

Tier 3: Confidential

Sensitive data whose unauthorized disclosure would cause significant harm, regulatory penalty, or breach of contractual obligations. Access is need-to-know, not role-based.

Examples: customer PII, billing records, employee HR data, legal contracts, non-public grid reliability data

Access: explicit individual authorization required, not just team membership. Multi-factor authentication is enforced. Full audit logging on every access event.

Tier 4: Restricted

Highest sensitivity. Unauthorized access poses risk of infrastructure failure at a national scale — not just financial loss. Subject to mandatory NERC CIP controls. If SCADA configurations or equipment control data were accessed by a malicious actor, the consequence is potential grid attack and widespread power loss across AEP's entire service territory.

Examples: SCADA control system configurations, generation asset settings, NERC CIP-regulated systems, cybersecurity incident data, equipment control parameters

Access: named individuals only. Director-level approval required. End-to-end encryption in transit and at rest. Active intrusion detection and firewall monitoring. Air-gapped systems where mandated by NERC CIP.

KPI

MEASURE	TARGET	RATIONALE
Vulnerability assessments	Quarterly across all systems	NERC CIP requires regular assessment; quarterly frequency matches the pace of threat landscape changes
Security patch application	100% within 30 days of release	Unpatched systems are the most common cause for successful cyberattacks on critical infrastructure
Tier 4 access events logged	100% — zero unlogged access	NERC CIP CIP-007 mandates complete audit trails for all access to critical cyber assets

5. Data Management & Standardization

AEP's 6 operating companies and multiple business segments have historically maintained independent data systems, resulting in inconsistent terminology, duplicate records, and conflicting metrics that make enterprise-wide analysis unreliable. A customer record may exist simultaneously in multiple operating company systems, each with slightly different field names, formats, and values. Different subsidiaries may label and measure an outage event in different ways, which means that companywide reliability reporting relies on manual reconciliation instead of trusted data. Similarly, generation of output data, financial records, and asset maintenance histories face the same fragmentation problem across AEP's broader enterprise. All other governance principles rely on standardization; without it, the organization cannot sustainably achieve data quality, accessibility, and trust.

AEP will establish a centralized enterprise data dictionary that defines standard terminology, data formats, and business rules for all critical data domains across every business segment, including customer records, grid asset data, outage events, generation metrics, and financial transactions. This dictionary will serve as the authoritative reference for how data is defined and used across the entire organization, and all operating companies and business units will be required to align their local data structures to these enterprise standards. Data Architects will be responsible for enforcing compliance through technical validation controls that check incoming data against defined schemas before it is admitted into enterprise systems. Data Stewards will oversee adherence to standardization policies at the operating company level, escalating non-compliance to the enterprise governance body.

To eliminate duplication across all parts of the business, a system of record will be designated for each major data domain: a single centralized source from which all downstream systems and analytics platforms consume data rather than maintaining independent copies. This eliminates the risk of conflicting versions of the same data existing simultaneously across the organization. A self-service portal for data discovery will allow authorized users across all business segments to search available data assets, understand their definitions and source systems, and access governed data without requiring IT intervention. Together, these measures reduce shadow data practices, accelerate analytics delivery, and build the organizational confidence necessary for data-driven decision-making at enterprise scale.

KPI

MEASURE	TARGET	RATIONALE
Metadata completeness across governed domains	95% across all domains	Incomplete metadata is the root cause of the manual reconciliation problem the DGGP is designed to eliminate
Data dictionary coverage	100% of critical data domains documented	An enterprise standard only works if every domain is covered, partial coverage still enables fragmentation

6. Governed Critical Data as a Service

AEP's operational decisions across all business segments depend entirely on data that is accurate, timely, and complete. In the Transmission and Distribution segment, when a grid operator receives inaccurate sensor

readings, crews may be dispatched to the wrong location, wasting resources and delaying restoration for customers. In the VIU segment, erroneous billing data erodes customer trust and exposes AEP to regulatory complaints. In generation operations, inaccurate output data can affect energy market commitments and compliance reporting. Incomplete or incorrect data in any part of the enterprise carries operational, financial, and in some cases life-safety consequences that span AEP's entire service territory across 11 states.

AEP will implement a proactive monitoring model to continuously assess data quality before distribution, rather than relying on downstream users to report issues. Automated monitoring tools, analogous to sensors checking water quality before it reaches homes, will run quality checks across all critical data pipelines in real time, catching problems at the point of ingestion rather than after decisions have already been made. The teams responsible for building and maintaining these pipelines will also own the responsibility for monitoring and resolving quality issues, ensuring clear accountability for data integrity across all of AEP's business units, including Transmission, Distribution, Generation, VIU operating companies, and corporate functions.

Data quality performance will be tracked through a centralized real-time dashboard visible to Data Stewards and Data Owners across the entire enterprise, displaying key metrics including duplicate record rates, null value percentages, rejected record counts, cleansed record volumes, and data delivery timeliness. Quality thresholds will be defined for each critical data domain, and automated alerts will be triggered when metrics fall below acceptable levels, enabling governance teams across any part of the organization to intervene before degraded data reaches analytics platforms or operational systems. This transforms data quality from a reactive cleanup exercise into a managed, measurable discipline that all AEP stakeholders can rely on regardless of their business segment.

KPI

Targets differ by data domain because the consequence of failure differs. Billing errors affect individual customers and trigger regulatory complaints. SCADA errors affect grid operations in real time — the consequence is immediate and physical, not financial. The targets reflect that distinction.

DATA DOMAIN	ACCURACY TARGET	RATIONALE
Billing data	99% accuracy	At 5.6 million customers, 1% error = 56,000 incorrect bills. This is the threshold where FERC regulatory exposure becomes real
SCADA operational data	99.99% accuracy	SCADA drives real-time grid decisions. An incorrect sensor reading at the wrong moment can cascade into infrastructure failure affecting millions of customers. The consequence is immediate and physical, not recoverable in the next billing cycle
Data quality issue resolution	24-hour resolution SLA	Grid operations cannot wait. A quality issue that persists beyond 24 hours risks feeding degraded data into operational decision-making

7. Prefer Well-Managed Data from Trusted Sources

AEP integrates data from a wide range of external sources across all its business segments, including weather data providers, ISO and RTO energy market feeds, third-party asset management platforms, equipment vendors, and regulatory reporting systems. The quality of AEP's analytics and operational decisions across the entire enterprise is directly dependent on the quality of these inputs. Inaccurate weather forecasts can distort

demand planning in the generation segment; erroneous market data can affect energy trading decisions; and faulty asset data can misdirect maintenance crews in the field. Not all external providers maintain the same standards of accuracy, timeliness, and reliability, and consuming data from unvetted sources introduces errors that can propagate silently through AEP's enterprise systems before anyone detects a problem.

It will maintain a curated registry of approved external data sources, each evaluated against defined criteria before being authorized for use across any business segment. Preferred sources are those that adhere to relevant regulatory and compliance standards, have established reputations in the energy industry, and can demonstrate consistent data quality track records. Internal systems of record will always be the preferred source for governed data domains across all parts of the enterprise, with external sources used only where internal data is unavailable or insufficient. All external data will be validated at the point of ingestion through automated checks that detect anomalies, format inconsistencies, and outliers before the data enters any of AEP's enterprise systems.

To protect the company from the consequences of external data failures across any segment of its business, all approved data providers will be subject to formal data quality agreements that function similarly to supplier contracts. These agreements define expected service levels, acceptable error rates, delivery timeliness, and the provider's liability in the event of data quality failures. If a provider consistently delivers substandard data, AEP reserves the right to suspend the relationship, require remediation before resuming data consumption, and seek alternative sources. This approach ensures that AEP's preference for trusted, well-managed data is not merely a policy aspiration but a contractually enforced standard that holds all external partners accountable for the quality of what they deliver to the organization.

KPI

MEASURE	TARGET	RATIONALE
External sources in approved registry	100% before use in any system	An unapproved source used by one operating company without enterprise review reintroduces the fragmentation problem governance is designed to prevent
Data quality agreements in place	100% of approved external providers	A policy preference for trusted sources is unenforceable without contractual obligations that hold providers accountable

8. Non-Governed Data

A governance framework that is too rigid risks becoming a barrier to innovation across AEP's enterprise. Data scientists, engineers, and analysts working across all business segments need the freedom to experiment with new data sources, test machine learning models, and prototype analytical solutions. This work is exploratory by nature and does not yet meet the standards required for governed production systems. If every experimental dataset must pass through the full governance approval process before it can be touched, the pace of innovation slows dramatically, and AEP risks falling behind in developing the advanced analytics capabilities that are central to its long-term strategy across areas such as predictive grid maintenance, renewable energy forecasting, generation optimization, and AI-driven customer analytics.

To balance governance discipline with innovation freedom, AEP will maintain a dedicated non-governed data sandbox environment, which is a secure, isolated space that is separate from production systems and

governed data assets across all business segments. Working within the sandbox, authorized data professionals from any part of the organization can access experimental data sources, develop and test new models, and explore emerging analytical techniques without any risk of disrupting live operations or compromising governed data. The sandbox will itself be maintained with appropriate access controls and security standards, ensuring that the freedom to experiment does not come at the cost of exposing AEP to unnecessary security vulnerabilities.

When experimental work in the sandbox produces results of sufficient quality and business value, it will follow a formal promotion pathway to become a governed asset available across the enterprise. The data scientist or analyst will present their findings to relevant business and technical stakeholders for evaluation. The governance team will then assess the solution for compliance with AEP's data standards, security requirements, and data quality thresholds, with legal review conducted where regulatory implications exist. Only upon receiving full approval will the solution be migrated into the governed production environment. This structured pathway ensures that innovation is encouraged and rewarded across all of AEP's business segments, while the integrity of the governed data ecosystem is never compromised.

KPI

MEASURE	TARGET	RATIONALE
Direct production deployments without governance review	0%	The sandbox exists precisely to prevent experimental work from reaching production without scrutiny. Any bypass of the promotion pathway undermines the integrity of the governed environment
Promotion pathway review time	Within 10 business days of submission	A slow review process creates pressure to bypass governance. Speed of review is a governance health indicator

9. Controls and Communication

A data governance framework is only as effective as the communication structures that support it. At AEP, governance responsibilities are distributed across 6 operating companies, multiple business segments spanning generation, transmission, distribution, and corporate functions, and four distinct role types. This creates significant potential for ambiguity about who is responsible for data decisions, who must be informed when issues arise, and how governance policies are enforced consistently across the entire organization. Without deliberate communication architecture, accountability breaks down and governance becomes inconsistent despite being well-documented in policy.

When a data quality issue is identified anywhere across AEP's enterprise, for example, a Data Owner in one of the VIU operating companies discovering corrupted meter readings, communication must follow a structured and need-to-know pathway. The issue is escalated to the enterprise governance body and the CDO, while Data Architects and technical teams are immediately engaged to contain and resolve it. Data users whose work is directly affected are notified promptly with a calm, factual message that acknowledges the issue and provides an estimated resolution timeline, preventing uninformed speculation while avoiding unnecessary alarm across the broader organization. Not every employee across AEP's 11 states needs to know about every data issue; communication is targeted to those who are directly impacted or directly responsible for resolution, maintaining operational calm while ensuring clear accountability.

All governance decisions, issue resolutions, policy updates, and meeting outcomes across the entire enterprise will be stored in a centralized governance portal, which is a searchable and access-controlled repository that serves as AEP's authoritative record of all governance activity. This portal ensures that decisions made today remain traceable and auditable years later, supporting both internal accountability and the external regulatory reviews that AEP is subject to as a federally regulated utility. Separation of duties will be enforced across all four roles, ensuring that no single individual has unchecked authority over data policy definition, technical implementation, and data usage simultaneously, regardless of which business segment or operating company they belong to.

KPI

MEASURE	TARGET	RATIONALE
Data quality issue escalation and resolution	24-hour resolution SLA	Unresolved issues feed degraded data into operational systems; 24 hours is the maximum tolerable window before grid or customer impact becomes likely
Governance portal documentation completeness	100% of decisions and resolutions logged	An undocumented governance decision cannot be audited and AEP is subject to external regulatory audit across all 11 states

10. Data Education

The success of AEP's data governance framework ultimately depends on the people who interact with data every day across all 11 states and business segments. Even the most sophisticated technical controls and well-written policies will fail if employees do not understand their data responsibilities, the importance of data quality, or how to handle sensitive information correctly. A sustained and role-tailored investment in data education is therefore essential to making governance a living organizational practice rather than a document that exists on paper without meaningful adoption across the workforce.

AEP will implement a tiered data education program structured around the distinct needs of different employee populations across the entire enterprise. Field technicians and operational staff across Transmission, Distribution, and Generation will receive foundational training focused on data handling responsibilities, covering what data they are accountable for capturing, how to capture it accurately, what constitutes sensitive information, and what they are not permitted to share or access. This baseline training ensures that the people closest to AEP's physical operations understand their role in maintaining data integrity at the source. Data scientists, analysts, and technical staff across all business segments will receive advanced training covering governance-compliant experimentation within the sandbox environment, data quality standards, pipeline management, and the formal promotion pathway for moving experimental work into governed production systems. Executives and senior leaders across the enterprise will receive training focused on accurately interpreting enterprise data dashboards, understanding governance accountability structures, and fulfilling their responsibility to champion a data-driven culture from the top of the organization downward.

All training will be delivered through a combination of in-person workshops and e-learning modules, with completion tracked through AEP's learning management system to ensure full organizational coverage across all locations and business units. Data governance training will be included in onboarding for all new employees

and refreshed annually for current staff to incorporate policy updates, regulatory changes, and lessons from data incidents. By building a workforce that understands and values data governance at every level, from field crews to the boardroom and across every part of AEP's operations, the organization creates the cultural foundation necessary for its governance framework to deliver sustained, long-term value enterprise wide.

MEASURE	TARGET	RATIONALE
Annual training completion	100% of all employees	An enterprise standard only works if every employee across all six operating companies, and 11 states understands their data responsibilities. Partial coverage means partial governance
Onboarding training completion	Within 30 days of start date	New employees handle data from day one. A 30-day window balances the competing demands of onboarding across a large, geographically distributed organization while closing the governance gap as quickly as possible

Conclusion

The ten principles in this document are not independent policies; they form a coherent system where each principle depends on the others. Standardization makes accessibility meaningful, because consistent definitions are the precondition for confident access decisions. Critical data quality monitoring depends on trusted sources, because automated checks cannot compensate for inputs that are structurally unreliable. The sandbox environment only creates value if the pathway into the governed production environment is rigorous enough to maintain integrity.

Every control, every number, and every timeline in this framework is measured to the consequence of failure and not to what is convenient to implement. The 99.99% accuracy target for SCADA data exists because a grid error is immediate and physical. The 24-hour access revocation window exists because a former employee retaining access to Tier 4 restricted data is a measurable risk, not a theoretical one. The 30-day onboarding training window exists because an untrained employee handling data from day one is a governance gap regardless of competing priorities.

AEP's \$54 billion capital investment plan will continuously introduce new data sources, new regulatory obligations, and new governance challenges. This framework was designed to absorb that change through monthly reviews, automated monitoring, and a centralized governance portal. The ultimate measure of its success is not compliance with its ten principles. It is whether AEP's 5.6 million customers receive reliable service, whether regulators can trust AEP's reporting, and whether the organization can make confident decisions based on data it genuinely trusts.